



Unit – 2

Demand Analysis

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1. INTRODUCTION

In the previous unit, we learnt about the scope and importance of managerial economics. We learnt that managerial economics helps managers to arrive at decisions that effectively use the firm's resources, and thereby lead the firm to grow profitably. In this unit, we will study demand analysis. The sustenance and growth of a business firm are greatly influenced by the demand for a firm's products (goods and/or services). In this unit, we shall explore the importance of demand business decisions/processes like pricing and forecasting. We begin our in-depth understanding of the subject with a foundation on 'demand'. Demand and supply are the two main concepts in economics. Some experts are of the opinion that the entire subject of economics can be summarised in terms of these two basic concepts.

1.1 Learning Outcome

After studying this unit, you should be able to:

- Describe the concept of demand and its features.
- Define and interpret the demand schedule, law of demand and price- quantity relationships and exceptions to the law of demand.
- Categorize the various factors which influence the demand for goods and services.
- Apply the concept of elasticity of demand and different kinds of elasticity of demand.



2. MEANING AND LAW OF DEMAND

In this section, we will discuss the meaning of demand and understand the law of demand. The term demand is different from desire, want, will or wish. In economics, demand has different meanings. Any want or desire will not constitute demand.

Demand = Desire to buy + Ability to pay + Willingness to pay + at a Point of Time

The term 'Demand' refers to the total or given quantity of a commodity or a service that is purchased by the consumer in the market at a particular price and at a particular time.

The following are some of the key features of demand:

- It is backed up by adequate purchasing power.
- It is always at a price.
- It should always be expressed in terms of specific quantity.
- It is related to time.

Consumers create the demand for a good or service based on the utility they perceived from them. There is a direct relation between the two i.e., higher the utility, higher would be demand and lower the utility, lower would be the demand.

(**Note:** Utility is the total amount of satisfaction derived by a consumer after consuming a product)

Definitions of Demand:

According to Melvin and Boyes, "Demand is a relationship between two variables, price and quantity demanded with all other factors that could affect demand being held constant.

According to Ferguson, "Demand refers to the quantities of a commodity that the consumers are able and willing to buy at each possible price during a given period of time, other things being equal."

2.1. Types of Demand

1. **Individual Demand:** refers to the quantities of a commodity that an individual consumer is willing to purchase at various prices during a given period of time. An Individual consumer is

called as a household in economics. For instance, the quantity of milk purchased per day by a household is the individual demand for milk.

2. **Market Demand:** refers to the total quantities of a commodity that all households are willing to buy at various prices during a given period of time. For instance, the total quantity of milk which all the buyers are willing to buy at a given price per day or over a week or month is the market demand for milk.
3. **Related Demand:** refers to the demand for two or more products which are jointly demanded together. For Example, Car and Petrol, Bread and Jam.
4. **Derived Demand:** Derived demand refers to the demand for a good or service that arises not for its own sake but as a result of the demand for another good or service. It is a fundamental concept in managerial economics, particularly in the context of production, supply chain management, and resource allocation. Derived demand occurs because the value of a good or service depends on its contribution to producing a final product that consumers desire. For example, the demand for raw materials, machinery, or labour in an industry is derived from the demand for the final goods or services produced using those inputs. For example, Consider the automobile industry. The demand for steel, a key raw material used in car manufacturing, is not independent. It is derived from the demand for cars. If consumer demand for cars increases, manufacturers will require more steel to produce more vehicles, leading to an increase in the demand for steel. Conversely, a decline in car sales will reduce the demand for steel.
5. Similarly, the demand for auto parts, such as tyres and engines, is also derived from the demand for cars.
6. **Composite demand:** Demand for goods that have multiple uses is called composite demand. A commodity is said to have composite demand when it can be put to multiple uses. For instance, from steel we can manufacture Utensils, cars, auto mobile bodies, grills, spare parts etc.

2.2. Determinants of Demand

The factors that affect or influence demand for a commodity or service are called the 'Determinants of demand'.

The **demand function** represents the relationship between the quantity of a product demanded and the factors influencing it. It is a mathematical expression that shows how changes in these factors affect the demand for a good or service. The general form of the demand function is:

$$Q_d = f(P, I, P_s, T, E, N)$$

Where:

- Qd: Quantity demanded
- P: Price of the good
- I: Consumer Income
- Ps: Prices of substitute or complementary goods
- T: Consumer tastes and preferences
- E: Consumer expectations about future prices or income
- N: Number of consumers in the market

Key Insights

Price is the primary determinant: As the price of a product rises, demand typically decreases (law of demand).

Income affects demand: Normal goods see increased demand with rising incomes, while inferior goods experience a decline.

Substitute and Complementary Goods: An increase in the price of a substitute good raises demand, while the price of a complementary good affects demand inversely.

For example, If the price of coffee increases, its demand may decrease (movement along the curve), while demand for tea, a substitute, might rise (shift in the curve).

The demand determinants in brief are as follows:

1. Price of the given commodity.
2. Prices of other substitute and/or complement related goods,
3. Future expected trend in prices.
4. Level of income and living standards of the people.
5. Tastes, preferences, customs, habits, fashion, and styles.
6. Advertisements.

7. Weather and climatic conditions.
8. Size, rate of growth and composition of population.
9. Government policy.

Thus, several factors are responsible for bringing changes in the demand for a product onto the market. A business executive should have the knowledge and information about all these factors and forces to finalize his own production, marketing, and other business strategies.

Price: Price is the basic factor on the basis of which a consumer usually decides to buy goods or services. When the prices are low, the quantity demanded is more. Similarly, When the price is high, the quantity demanded is low and consumers purchase very less quantities. The relationship is expressed as $D=f(P)$

In this equation D represents demand which is the dependent variable and P is the price which is the independent variable.

Income: The purchasing capacity is determined by the buyer's income. Demand for a commodity depends on the buyer's income. With the buyer's income increases, more goods, products and commodities are purchased. Demand for luxurious and expensive goods is related to the income of the buyers.

Taste, habits and preferences: Demand for various goods depends on the consumers' taste and preferences. For instance, fast food, snacks, ice creams, sweets and beverages completely depend on individual taste and preference. Whereas the demand for Cigarettes, Alcohol and Tobacco depends on the habits of the consumers. Vegetarians may have no preference towards Non veg or meat-based products. Similarly, a Non-Vegetarian will demand for meat-based products even when the price is very high. Taste, preference and habits change from time to time and similarly demand for the related products changes too.

Complementary and Substitutes: Complementary goods are goods that are used with other products to satisfy a particular demand. For example, Petrol is a complementary product used to ride a bike. Therefore, in order to satisfy the consumer demand for bikes, complementary products like petrol are needed. As we cannot ride a bike without petrol unless the bike is an electric vehicle.

On the other hand, Substitute are goods that are used as an alternative or instead of the actual product demanded. For example, if the price of coffee goes up tea can be used instead of coffee. If the price of

coffee increases the demand for coffee will go down. In this case customers will switch to Tea if the price of Tea is less than Coffee.

Consumer's Expectations: When a consumer expects a future change in price then the demand for that product will change. When the consumer expects a decrease in future prices of a product the demand decreases. This is because the consumer buys a smaller quantity of the commodity as the prices may fall in future. Similarly, if the consumer expects an increase in the future price of a product, then the demand increases. This is because the consumer buys more currently due to the expectation that the prices may go up and the product may become costly. For instance, demand for daily commodities in India like tomato and onions. Homemakers with limited income may buy fewer quantities of onions if they expect a fall in price in future and may buy or demand large quantities if they expect the price to increase in future.

Advertisements: Consumer preferences can change due to the influence of advertisements, sales promotions and offers to a certain extent. Consumer preferences for fast moving consumer goods such as soaps, shampoos toothpaste, cosmetics and detergents are largely influenced by advertising.

Consumer Credit Facilities: If consumers are able to get loans and credit facilities easily, they will be tempted to make more purchases of certain commodities thereby increasing the demand for such commodities.

Government Policy: Economic Policies of the government influence the demand for commodities. If the government imposes taxes on various commodities in the form of GST, import export duties etc. The price of these commodities will increase. When the price increases the demand for these commodities will decrease. When the government regulates the prices of certain commodities the price reduces and the demand increases. On the other hand, if the government incurs expenditure on construction of roads, flyovers or any form of infrastructure then, the demand for goods related to construction and technology will increase.

Weather and Climatic conditions: Demand for various types of goods depends on the climatic factors as different goods are needed during different climates. During summer demand for cold juices, ice cream and cotton clothes increases. During winter there is a great demand for warm clothes, moisturizers, hot beverages, heaters.

Size and composition of the population: Market demand for a commodity depends on the size and composition of the population. The population of a country determines the number of consumers and

type of consumers. The larger the population of a country is, the greater the number of consumers a business can tap. An increase in the size of the population will increase the demand for a commodity and vice versa. Composition of the population like children, teenagers, adults, old people, male and female decide the type of consumers that influence demand of various commodities that are used by them. For instance, if there are a greater number of old people demand for medical facilities, health care insurance will be more. If the population of babies is considerably large then demand for baby products, baby strollers, paediatric facilities will be high.

2.3. Demand Schedule

Individual Demand Schedule

The individual demand schedule is the table which shows various quantities of a commodity that would be purchased at different prices by a household during a given period. The demand schedule explains the functional relationship between price and quantity. It is a list of various amounts of a commodity that a consumer is willing to buy (and so seller to sell) at different prices at a particular period of time.

Table 1: Individual Demand Schedule for Apples

Price (Rs. Per Kg)	Quantity demanded in Units (Kg per Week)
100	1
90	2
80	4
70	6

Source: *Economics by DK Sethi and U Andrews*

Table 1, Represents the individual demand schedule for apples. The table shows the quantity demanded by an individual person per week. When the price falls from Rs100 to Rs.90 the quantity demanded also increases from 1 kg to 2 kgs and so on.

Market demand schedule

The market demand schedule is a table which shows various quantities of a commodity that all the buyers or consumers are willing to purchase at different prices during a given period. When the demand schedules of all buyers are taken together, we get the aggregate or market demand schedule. In other words, the total quantity of a commodity demanded at different prices in a market by the whole body of consumers at a particular period of time is called a market demand schedule. It refers to the aggregate behaviour of the entire market rather than mere totalling of individual demand schedules. It is the horizontal summation of the individual demand schedule. The market demand schedule is more continuous and smoother when compared to an individual demand schedule. Table 2 gives an example of a market demand schedule.

Table 2: Market Demand Schedule for Apples

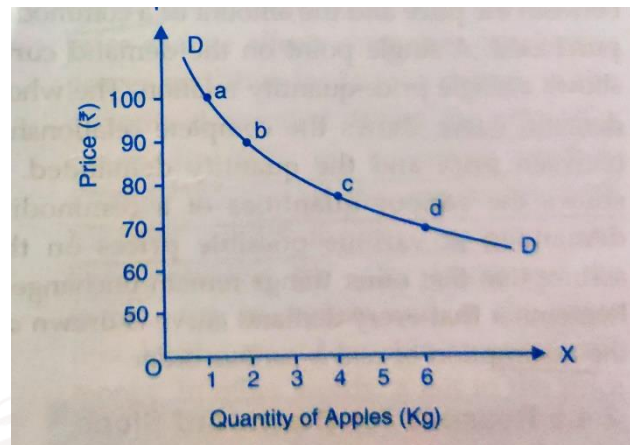
Price (Rs. Per Kg)	Quantity demanded by person A (Kg per week)	Quantity demanded by person B (Kg per week)	Total MarketDemand (A + B) (Kg per week)
100	1	2	1 + 2 = 3
90	2	3	2 + 3 = 5
80	4	5	4 + 5 = 9
70	6	7	6 + 7 = 13

Source: *Economics by DK Sethi and U Andrews*

Table 2 represents the Market demand schedule for Apples. Let us assume that A and B are the only individuals buying apples in the market. The total market demand is derived by adding the individual demand of Person A and Person B. The table shows that as the price falls from Rs100 to 90 the Total market demand increases from 3 to 5 Kgs per week. This shows that the total quantity demanded falls when there is an increase in price and the total demand increases when the price falls.

Demand curve

A demand curve is a locus of points showing various alternative price- quantity combinations. In short, the graphical presentation of the demand schedule is called a demand curve. Figure 1 depicts the demand curve.



Source: Economics by DK Sethi and U Andrews

The demand curve shown in figure 1 represents 'DD', a demand curve or an individual demand curve. X axis represents quantity demanded and Y axis represents the price. The graph shows the functional relationship between quantity demanded and prices of a given commodity. The demand curve has a negative slope or it slopes downwards to the right. The negative slope of the demand curve clearly indicates that quantity demanded goes on increasing as price falls and vice versa.

The market demand curve is a curve that represents different quantities of goods which all the consumers in the market are willing to buy at different prices during a specified period. The market demand curve can be drawn by aggregating together all the individual demand curves. It is the horizontal summation of the demand curves of all the households.

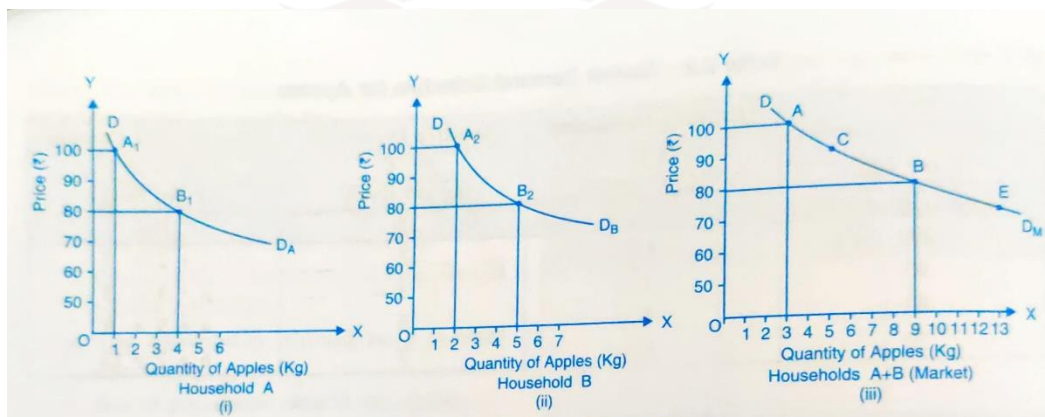


Fig 2 Individual and market demand curve

Source: Economics by DK Sethi and U Andrews

Table 3: Market Demand Schedule for Apples

Price (Rs. Per Kg)	Quantity demanded by person A (Kg per week)	Quantity demanded by person B (Kg per week)	Total Market Demand (A + B) (Kg per week)
100	1	2	1 + 2 = 3
90	2	3	2 + 3 = 5
80	4	5	4 + 5 = 9
70	6	7	6 + 7 = 13

Source: Economics by DK Sethi and U Andrews

Figure 2 shows the individual market demand curve on panel (i) and (ii) which is represented as 'DDA' and 'DDB' respectively. The graphs show the individual demands of Person A and Person B. The X-Axis represents the number of apples demanded, and Y-Axis represents the price per Kg. The Market demand curve is represented as 'DDM', which slopes downwards, showing an inverse relationship between Price and Quantity demanded. We consider the price and Total market demand (A+B) from Table 3 to plot the graph. When the Price is Rs100 per kg the quantity of market demand is 3 and when the price goes down to Rs90 per Kg the total quantity demanded or the total market demand is 5. This shows that the demand increases as the price decreases.

Law of Demand

The **Law of Demand** is a fundamental principle in economics that describes the inverse relationship between the price of a good or service and the quantity demanded, assuming all other factors remain constant (**ceteris paribus**). It states that, as the price of a good increases, the quantity demanded decreases, and conversely, as the price decreases, the quantity demanded increases.

This relationship can be expressed as:

$$Q_d = f(P)$$

Where:

- Q_d : Quantity demanded
- P : Price of the good

The **Law of Demand** in economics explains the inverse relationship between the price of a good and the quantity demanded, assuming all other factors remain constant. It states that as the price of a good rises, the quantity demanded decreases, and as the price falls, demand increases. This occurs due to the **income effect**, where lower prices enhance consumers' purchasing power; the **substitution effect**, where consumers switch to cheaper alternatives; and **diminishing marginal utility**, which reduces the willingness to pay for additional units at higher prices. However, exceptions exist, such as **Giffen goods**, where higher prices can increase demand due to the income effect, and **Veblen goods**, where higher prices add prestige, making them more desirable. For example, if the price of bread drops, households may buy more, but if it rises, they may reduce consumption or switch to alternatives. Understanding this principle is critical in managerial economics for pricing, market forecasting, and designing strategies to influence consumer behaviour effectively.

Reasons for the downward slope of the demand curve

1. **Law of diminishing marginal utility:** The satisfaction derived from the consumption of successive units goes on falling, because earlier units have partly satisfied our wants. As we consume every additional unit of the commodity the satisfaction derived is less. A consumer wants to pay a lower price for additional units and only purchases when the price falls. Therefore, the demand curve slopes downward.
2. **Multiple uses of goods:** If the price of the goods falls, consumers use more of those goods for different purposes and the quantity demanded increases.
3. **Substitution effect:** When the price of any substitute goods falls, the consumer gives up the dearer goods or costly goods and buys additional units of the cheaper goods. In the same way when price falls, the consumers who are consuming the other goods are in turn attracted to the cheaper goods and this causes the demand curve to slope downward.
4. **Income effect:** When the price of a commodity falls, the real income of the consumer increases and the purchasing power of the consumer increases. This enables the consumer to buy more units and vice versa. This inverse relationship between quantity demanded and price causes the demand curve to slope negatively or downwards.

2.4. The Significant features of Law of Demand

The law of demand explains the relationship between price and quantity demanded of a commodity. It says that demand varies inversely with the price. The law can be explained in the following manner, “Keeping other factors that affect demand constant, a fall in price of a product leads to increase in quantity demanded and a rise in price leads to decrease in quantity demanded for the product”. The law can be expressed in mathematical terms as “Demand is a function of price”. Thus, symbolically $D = F(p)$ where, D represents Demand, P stands for Price and F denotes the Functional relationship. The law explains the cause-and-effect relationship between the independent variable [price] and the dependent variable [demand]. The law explains only the general tendency of consumers when buying a product. A consumer would buy more when price falls due to the following reasons:

1. A product becomes cheaper [Price effect]
2. Purchasing power of a consumer would go up [Income effect]
3. Consumers can save some amount of money
4. Cheaper products are substituted for costly products [substitution effect]

Assumptions of the law of demand

1. Price is an independent variable and demand is a dependent variable.
2. The operation of the law is conditioned by the phrase “Other things being equal” or “other things remain constant”. It indicates that given certain conditions, certain results would follow. The inverse relationship between price and demand would be valid only when tastes and preferences, customs and habits of consumers, prices of related goods, and income of consumers would remain constant.

2.5. Exceptions to the law of demand

Customers would buy more when the price falls in accordance with the law of demand. Exceptions to the law of demand state that, “with a fall in price, demand also falls, and with a rise in price demand also rises”. This can be represented by an upward-sloping demand curve. In other words, the demand curve slopes upwards from left to right.

Figure 3 depicts the exceptional demand curve. It is clear that as price rises from P1 to P2, the quantity demanded also expands from X1 to X2 units. The curve is an upward or positive sloping curve. Here, the law of demand does not work and the opposite happens here.

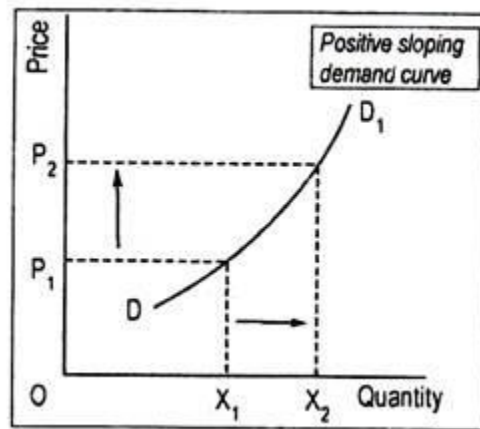


Figure 3: Exceptional Demand Curve

Some examples that favour the unusual demand curve are as follows:

1. **Giffen's paradox** – A paradox is an inconsistency or contrary. Sir Robert Giffen, an Irish Economist, with the help of example (inferior goods) disproved the law of demand. Giffen's paradox holds that "Demand is strengthened with a rise in price or weakened with a fall in price". He gave the example of the poor people of Ireland who were using potatoes and meat as daily food articles. When the price of potatoes declined, customers instead of buying larger quantities of potatoes started buying more meat (superior goods). Thus, the demand for potatoes declined in spite of a fall in its price.
2. **Veblen's effect** – It shows the quality-price relationship as sometimes consumers assume that high-priced goods are of better or superior quality. Thorstein Veblen, a noted American economist contends that there are certain commodities that are purchased by rich people not for their direct satisfaction, but for their 'snob-appeal' or 'ostentation'. Veblen's effect states that demand for status symbol goods would go up with a rise in price and vice versa. In the case of such status symbol commodities, it is not the price which is important but the prestige conferred by that commodity on a person makes him go for it. More commonly cited examples of such goods are diamonds and precious stones, Luxury cars, world famous paintings, commodities used by world famous personalities, etc. Therefore, commodities having 'snob-appeal' are to be considered as exceptions to the law of demand.
3. **Fear of shortage** – When serious shortages are anticipated by the people, (e.g., during the war period) they purchase a greater quantity of goods even though the current price is higher.

4. **Fear of future rise in price** – If people expect a future hike in prices, they buy more even though they feel that current prices are higher. Otherwise, they have to pay a still high price for the same product.
5. **Speculation** – Speculation implies purchase or sale of an asset with the hope that its price may rise or fall and make speculative profit. Normally, speculation is witnessed in the stock exchange market. People buy more shares only when their prices show a rising trend. This is because they make more profit if they sell their shares when the prices actually rise. Thus, speculation becomes an exception to the law of demand.
6. **Conspicuous consumption** – when consumers purchase some items although the items' prices are rising on account of their special uses in the modern style of life. For Example, Diamonds, Expensive phones and watches.
7. **Emergencies** – During emergency periods like war, famine, floods, cyclones, accidents, etc., people buy certain articles even though the prices are quite high.
8. **Ignorance** – Sometimes people may not be aware of the prices prevailing in the market. Hence, they buy more at higher prices because of sheer ignorance.
9. **Necessaries** – Necessaries are those items which are purchased by consumers whatever the price may be. Consumers would buy more necessities in spite of their higher prices.

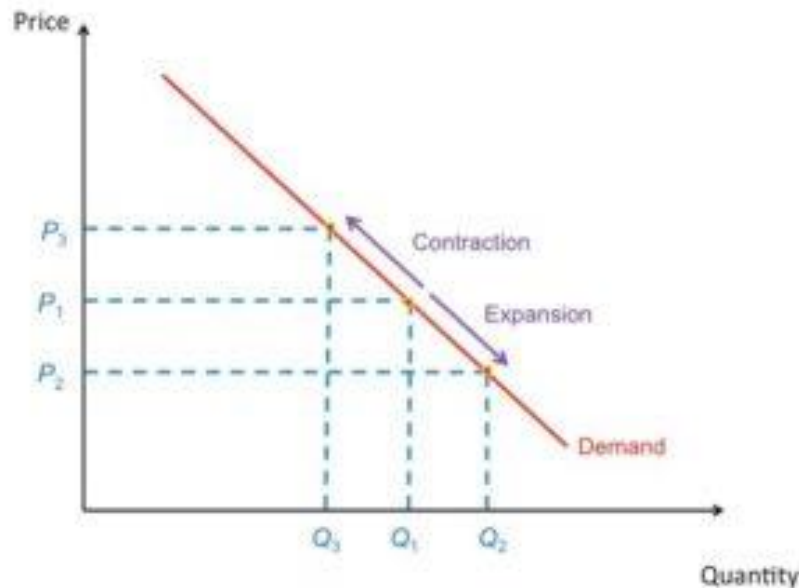
2.6. Changes in Quantity Demanded, Movement along the Demand Curve

When the quantity demanded of a commodity change (rises or falls) as a result of change in its own price, while other determinants of demand (like income, taste, prices of related goods like complementary and substitutes) remain constant. It is known as a change in quantity demanded.

1. Expansion of Demand
 2. Contraction of Demand
-
1. Expansion of Demand: When the quantity demanded of a commodity rises due to a fall in its price, other things remain constant. It is called a Rise in quantity demanded or Expansion of Demand. In fig 4, When the price falls from P3 to P1 the Quantity increases from Q3 to Q1 and so on. A movement down a demand curve is called a 'Rise in the Quantity demanded or expansion of demand'
 2. Contraction of Demand: When the quantity demanded of a commodity falls due to a rise in its price, other things remain constant. It is called Fall in quantity demanded or Contraction of

Demand. In fig 4, When the price rises from P_2 to P_3 the Quantity decreases from Q_2 to Q_3 and so on. A movement up the demand curve is called a 'Fall in the Quantity demanded or Contraction of demand'.

**Fig 4 Graph showing Movement along demand Curve,
Expansion and Contraction of Demand**



Source: Economics by DK Sethi.

2.7. Changes in Demand, Shifts in Demand Curve

When the amount purchased of a commodity rises or falls because of changes in factors other than the own price of the commodity it is called a change in demand.

1. Increase in Demand
2. Decrease in Demand

1. **Increase in demand:** It refers to a situation when consumers buy larger quantities of a commodity at the same price due to changes in factors other than the own price of the commodity. A rightward shift in the demand curve indicates an increase in demand. As per fig 5 Price is the same but the quantity increases from Q to Q_1 due to changes in other factors. The demand curve shifts towards the right.
2. **Decrease in demand:** Refers to a situation when consumers buy smaller quantities of a commodity at the same price due to changes in factors other than the own price of the commodity.

When the demand curve shifts to the left it indicates a decrease in demand. As per fig 5 Price is the same but the quantity decreases from Q to Q_2 due to changes in other factors. The demand curve shifts towards the left.

Fig 5 Graph showing increase and decrease in demand

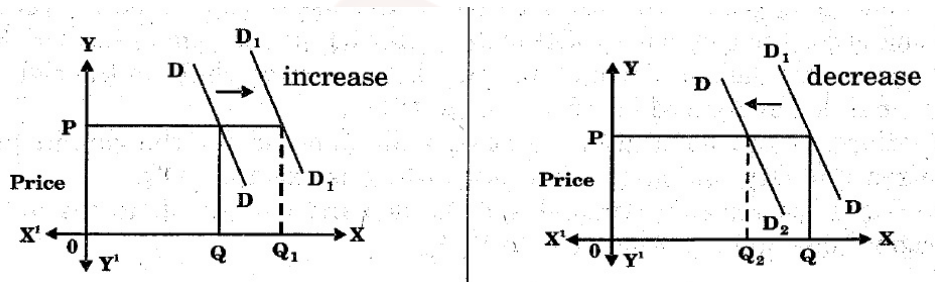


Table 4 showing Change in Quantity demanded - movement along the demand curve

Expansion of Demand	Rise in demand due to fall in Price while other things remain constant.	Downward movement of curve	Change in demand due to Price.
Contraction of Demand	Fall in demand due to rise in Price while other things remain constant.	Upward movement of curve	Change in demand due to Price.

Table 5 showing Change in Demand - Shift in demand curve

Increase in Demand	Rise in demand due to change in other determinants or factors.	Rightward shift of demand curve	Price remains constant. Change in demand due to change in other factors
Decrease in Demand	Fall in demand due to change in other determinants or factors.	Left ward shift of demand curve.	Price remains constant. Change in demand due to change in other factors

Source: Economics by Dk Sethi

SELF-ASSESSMENT QUESTIONS – 1

Fill in the blanks:	
1	In a typical demand schedule quantity demanded varies _____ with price.
2	In the case of Giffen's goods, price and demand go in the _____ directions.
3	If demand changes as a result of price changes, then it is a case of _____ and _____ in demand
4	Law of demand is a _____ statement.
5	Demand function is much more than law of demand.
6	In the case of Veblen goods, a fall in price leads to a _____ in demand.
True or False Questions	
i)	Ogburn and Nimkoff defined "Sociology is the scientific study of social life."
ii)	The scope of sociology can be defined as the approach it takes to study a particular field, not just what it studies.
iii)	According to Spencer, sociology attempts to understand or interpret social action
iv)	Weber coined the term "sociology"
v)	Enlightenment is the movement towards religious and supernatural beliefs.



3. ELASTICITY OF DEMAND

Earlier we discussed the law of demand and its determinants. It tells us only the direction of change in price and quantity demanded. But it does not specify how much more is purchased when price falls or how much less is bought when price rises. In order to understand the quantitative changes or rate of changes in price and demand, we have to study the concept of elasticity of demand. In this section, we will discuss the elasticity of demand.

Meaning and definition

The term elasticity is borrowed from physics. It shows the reaction of one variable with respect to a change in other variables on which it is dependent. Elasticity is an index of reaction.

In economics the term elasticity refers to a ratio of the relative changes in two quantities. It measures the responsiveness of one variable to the changes in another variable.

Elasticity of demand is generally defined as the responsiveness or sensitivity of demand to a given change in the price or determinant of a commodity. It refers to the capacity of demand either to stretch or shrink to a given change in price or non-price determinant.

Elasticity of demand = (Percentage change in quantity demanded)/(Percentage change in determinant of demand)

For e.g., The quantity demanded for a good/service changes by some percentage due to change in consumer income by some percentage, Measurement of these changes can lead to calculation of elasticity of demand. Elasticity of demand indicates a ratio of relative changes in two quantities, i.e., price and demand.

According to professor Boulding, 'Elasticity of demand measures the responsiveness of demand to changes in price'.

In the words of Marshall, "The elasticity (or responsiveness) of demand in a market is great or small, according to the amount demanded much or little for a given fall in price and diminishes much or little for a given rise in price"².

3.1. Kinds of elasticity of demand

Broadly speaking, there are five kinds of elasticity of demand.

They are price elasticity, income elasticity, cross elasticity, promotional elasticity and substitution elasticity. We shall discuss each one of them in some detail.

Price elasticity of demand

Price elasticity of demand may be defined as the degree of responsiveness of the quantity demanded of a commodity in response to a change in its price. Price elasticity of demand refers to the ratio of the percentage change in the quantity demanded of a commodity to a given percentage in its price.

Thus

$$E_p = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

Where, E_p is price elasticity of demand

The following are the five degrees of price elasticity of demand:

1. **Perfectly elastic demand** – In this case, a very small change in price leads to an infinite change in demand. The demand curve is horizontal

line and parallel to OX axis. The numerical co-efficient of perfectly elastic demand is infinity ($E_D = \infty$). Figure 6 depicts the perfectly elastic demand curve.

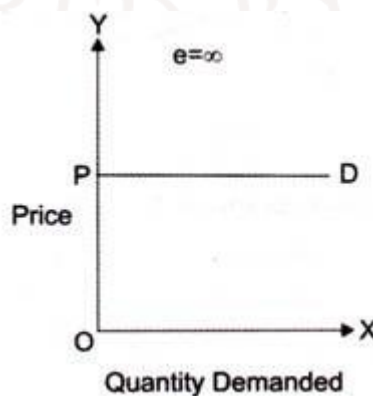


Figure 6: Perfectly Elastic Demand

Source: Economics for managers by Geetha M Rajaram

2. **Perfectly inelastic demand** – In case of any change in price, the quantity demanded will be perfectly constant. The demand curve is a vertical straight line and parallel to OY axis. It can be interpreted from Figure 7 that the movement in price from OP₁ to OP₂ and OP₂ to OP₃ does not show any change in the demand of a product (OQ). The demand remains constant for any value of price. Hence, the numerical co-efficient of perfectly inelastic demand is zero. $ED = 0$.

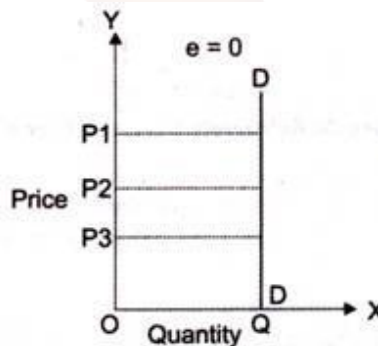


Figure 7 depicts perfectly inelastic demand curve.

Source: Economics for managers by Geetha M Rajaram

3. **Relatively elastic demand** – Here, if there is a small change in price, then it leads to a greater proportional change in demand. Figure 8 depicts the relatively elastic demand, here we see that the change in demand is more than that of change in price. Hence, the elasticity is greater than one. Hence, the numerical co-efficient of demand is greater than one. Relatively elastic demand is generally called as 'elastic demand' or 'more elastic' demand,

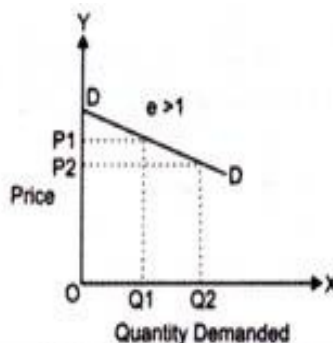


Figure 8: Relatively Elastic Demand

Source: Economics for managers by Geetha M Rajaram

4. **Relatively inelastic demand** – In this case, a large change in price leads to a less proportionate change in demand, this can be represented by a steeper demand curve. Hence, elasticity is less than one.

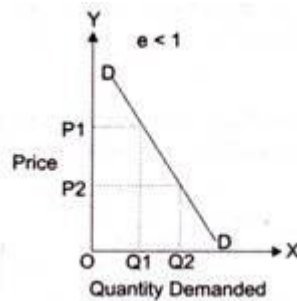


Figure 9: Relatively Inelastic Demand

Source: Economics for managers by Geetha M Rajaram

Figure 9 depicts the relatively inelastic demand curve. It can be interpreted that the proportionate change in demand from OQ_1 to OQ_2 is relatively smaller than the proportionate change in price from OP_1 to OP_2 . Relatively inelastic demand is popularly known as 'inelastic demand' or 'less elastic demand'.

5. **Unitary elastic demand** – Here, there is a proportionate change in price which leads to equal proportional change in demand. For e.g., 5 % fall in price leads to exactly 5 % increase in demand. Hence, elasticity is equal to unity. It is possible to come across unitary elastic demand but it is a rare phenomenon.

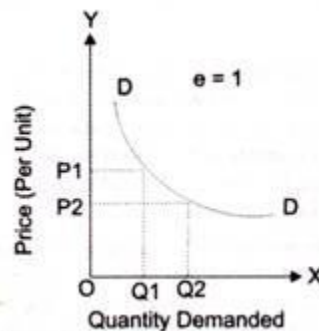


Figure 10 depicts the unitary elastic demand curve.

Source: Economics for managers by Geetha M Rajaram

From figure 10, it can be interpreted that change in price OP1 to OP2 produces the same change in demand from OQ1 to OQ2. Therefore, the demand is unitary elastic.

Out of five different degrees, the first two are theoretical and the last one is a rare possibility. Hence, in all our general discussions, we make reference only to two terms: relatively elastic demand and relatively inelastic demand.

3.2. Determinants of price elasticity of demand

You may observe that the elasticity of demand depends on several factors of which the following are some of the important ones:

1. **Nature of the commodity s**– Commodities coming under the category of necessities and essentials tend to be relatively inelastic, because the huge change in price of such goods could not bring any large change in their demand. For example, rice, wheat, sugar, milk, vegetables, etc.; On the other hand, for comforts and luxuries, demand tends to be relatively elastic, meaning small change in price may result in huge change in demand e.g., TV sets, refrigerators, etc.
2. **Existence of substitutes** – Substitute goods are those that are considered to be economically interchangeable by buyers. If a commodity has no substitutes in the market, demand tends to be relatively inelastic because people have to pay a higher price for such articles. For example, salt, onions, garlic, ginger, etc. In the case of commodities having different substitutes, demand tends to be relatively elastic. For example, blades, toothpaste, soaps, etc.
3. **Number of uses for the commodity** – Single-use goods are those which can be used for only one purpose and multiple-use goods can be used for a variety of purposes. If a commodity has only one use (single use product), demand tends to be relatively inelastic because people have to pay more prices if they have to use that product for only one use, for example, all kinds of eatables, seeds, fertilizers, pesticides, etc. On the contrary, for commodities having several uses, [multiple-use-products] demand tends to be relatively elastic, for example, coal, electricity, steel, etc.
4. **Possibility of postponing the use of a commodity** – In case there is no possibility of postponing the use of a commodity, demand tends to be relatively inelastic because people have to buy them irrespective of their prices, e.g., medicines. If there is a possibility of postponing the use of a commodity, demand tends to be relatively elastic, e.g., buying a TV set, motor cycle, washing machine, car, etc.

5. **Level of income of the people** – Generally speaking, demand will be relatively inelastic in the case of rich people, because any change in market price will not alter and affect their purchase plans. On the contrary, demand tends to be relatively elastic in case of poor.
6. **Durability and reparability of a commodity** – Durable goods are those which can be used for a long period of time. Demand tends to be relatively elastic in case of durable and repairable goods, because people do not buy them frequently, e.g., table, chair, vessels etc. On the other hand, for perishable and non-repairable goods, demand tends to be relatively inelastic e.g., milk, vegetables, electronic watches, etc.
7. **Range of prices** – There are certain goods or products like imported cars, computers, refrigerators, TV, etc. which are costly in nature. Similarly, a few other goods like nails, needles, etc. are low priced goods. In all these cases, a small fall or rise in prices will have an insignificant effect on their demand. Hence, demand for them is relatively inelastic in nature. However, commodities having normal prices are relatively elastic in nature.
8. **Range of prices** – There are certain goods or products like imported cars, computers, refrigerators, TV, etc. which are costly in nature. Similarly, a few other goods like nails, needles, etc. are low priced goods. In all these cases, a small fall or rise in prices will have an insignificant effect on their demand. Hence, demand for them is inelastic in nature. However, commodities having normal prices are relatively elastic in nature.
9. **Proportion of the expenditure on a commodity** – When the amount of money spent on buying a product is either too small or too big, demand tends to be relatively inelastic. For example, salt, newspapers or a site or house. On the other hand, if the amount of money spent is moderate, demand tends to be relatively elastic. For example, vegetables and fruits, clothes, provision items etc.
10. **Habits** – When people are used to a commodity or addicted, they do not care for price changes over a certain range e.g., in the case of smoking, drinking, use of tobacco, etc. In that case, demand tends to be relatively inelastic. If people are not used to the product, then demand generally tends to be relatively elastic.
11. **Period of time** – Price elasticity of demand varies with the length of the time period. Generally speaking, in a short period, demand is relatively inelastic because consumption habits of the people, customs and traditions, etc. do not change. On the contrary, demand tends to be relatively elastic in the long period where there is the possibility of all kinds of changes.

12. **Level of knowledge** – Demand in case of customers who are aware would be relatively elastic and in case of ignorant customers, it would be relatively inelastic.
13. **Existence of complementary goods** – Goods or services whose demands are interrelated such that an increase in the price of one of the products results in a fall in the demand for the other are known as complementary goods. Goods that are jointly demanded are relatively inelastic in nature. For example, pen and ink, vehicles and petrol, shoes and socks, etc. have inelastic demand for this reason. If a product does not have complementary goods, in that case demand tends to be relatively elastic. For example, biscuits, chocolates, ice creams, etc. In this case, the use of a product is not linked to any other product.
14. **Purchase frequency of a product** – If the frequency of purchase is very high, the demand tends to be relatively inelastic, e.g., coffee, tea, milk, match-box, etc. On the other hand, if people buy a product occasionally, demand tends to be relatively elastic, e.g., durable goods like radio, tape recorders, refrigerators, etc.

Thus, the demand for a product being elastic or inelastic will depend on a number of factors.

3.3. Measurement of price elasticity of demand

There are different methods to measure the price elasticity of demand and among them, the three most important methods are: total expenditure method, point method and arc method.

Let us discuss these methods in detail.

Total expenditure method

Elasticity of demand can be measured by considering the change in total expenditure incurred on a commodity as a result of change in the price of the commodity. Total expenditure incurred on a commodity before and after the price change is compared to understand the elasticity in demand.

The expenditure incurred by households on the purchase of a commodity is equal to the price of the product multiplied by the quantity demanded at that price.

$$TE = P \times Q$$

P is the price and Q is the quantity.

Or Total expenditure = Price per unit x Total quantity purchased

For example, if 10 units of a commodity are demanded when its price is Rs 6, the total expenditure will be Rs.60 ($10 \times 6 = 60$)

Under this method, the price elasticity is measured by comparing the total expenditure of the consumers (or total revenue i.e., total sales values from the point of view of the seller) before and after variations in price. Table 6 Table 2.3 shows the total expenditure of consumers with variations in price and quantity demanded. We measure price elasticity by examining the change in total expenditure as a result of a change in the price and quantity demanded for a commodity.

Total expenditure = Price per unit x Total quantity purchased

Table 6: Total Expenditure of Consumers

	Price in(Rs.)	Qty. Demanded	Total Expenditure	Nature of PED
I Case	5.00	2000	10000	> 1
	4.00	3000	12000	
	2.00	7000	14000	
II Case	5.00	2000	10000	= 1
	4.00	2500	10000	
	2.00	5000	10000	
III Case	5.00	2000	10000	< 1
	4.00	2200	8000	
	2.00	4200	8400	

Note:

Variation in the value of ED can be summarized as:

1. When a new outlay is greater than the original outlay, then $ED > 1$.
2. When new outlay is equal to the original outlay then $ED = 1$.
3. When new outlay is less than the original outlay then $ED < 1$.

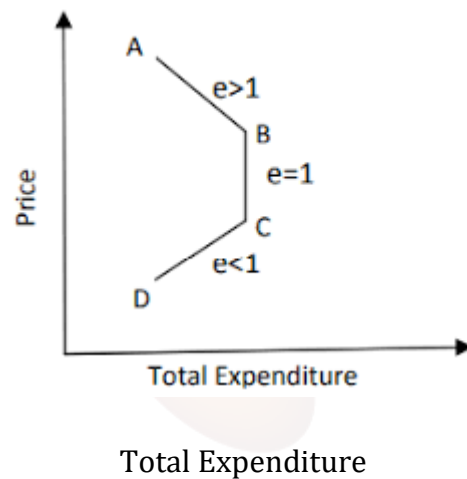


Figure 11: Graphical Representation of Total Expenditure Method

From the diagram it is clear that:

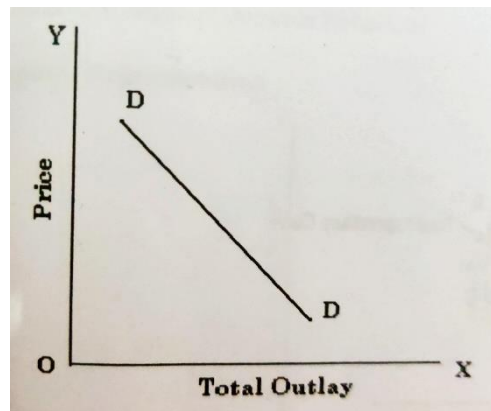
1. From A to B, price elasticity is greater than one.
2. From B to C, price elasticity is equal to one.
3. From C to D, price elasticity is less than one.

Note:

The following points should be noted from the total expenditure method:

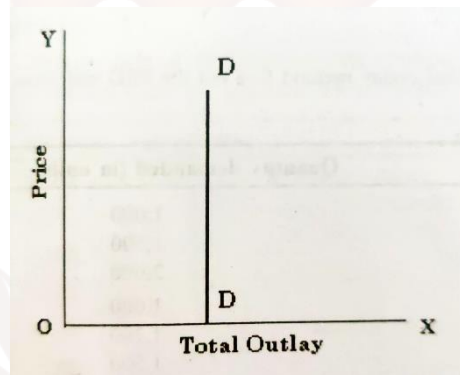
- When total expenditure increases with the fall in price and decreases with a rise in price, then the PED is greater than one.
- When the total expenditure remains the same either due to a rise or fall in price, the PED is equal to one.
- When total expenditure decreases with a fall in price and increases with a rise in price, PED is less than one.

Graph 1 - Relatively Elastic Demand - Total outlay method



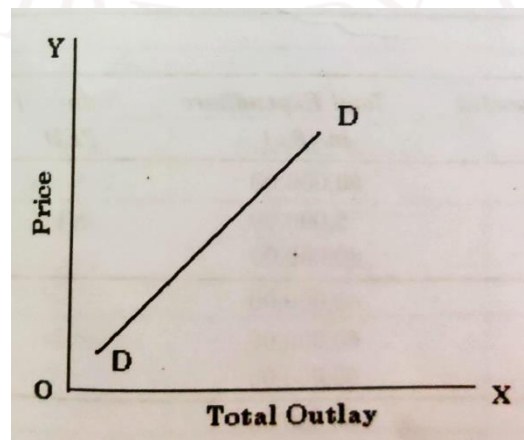
As per the graph when the new outlay (expenditure) is more than the original outlay, then price elasticity is greater than. I.e., $P_e > 1$

Graph 2 – Unitary Elastic - Total outlay method



As per the graph when the new outlay (expenditure) remains the same like the original outlay then the price elasticity is equal to one, I.e., $P_e = 1$

Graph 3 – Inelastic demand - Total outlay method



As per the graph, when the new outlay (expenditure) is less than the original outlay, then price elasticity is less than one. $P_e < 1$.

Point method

Prof. Marshall advocated this method. The point method measures price elasticity of demand at different points on a demand curve. This method is used when there is an insignificant change in quantity due to a very small change in price. Hence, in this case, an attempt is made to measure small changes in both price and demand. It can be explained either with the help of mathematical calculation or with the help of a diagram or graphical representation. Elasticity of demand is different on different points on the same demand curve. The price elasticity of demand at any point on the demand curve can be measured by:

$$e_p = \frac{\text{Lower segment}}{\text{Upper segment}}$$

1. On a straight-line demand curve:

To measure price elasticity of demand at point R on a linear or straight-line demand curve AB, which is intercepted by both axes. (Curve touches both Y and X axis). The lower line segment below point R is called RB and the upper line segment is RA

$$e_p \text{ at point R} = \frac{\text{Lower segment}}{\text{Upper segment}}$$

$$= -\frac{RB}{RA}$$

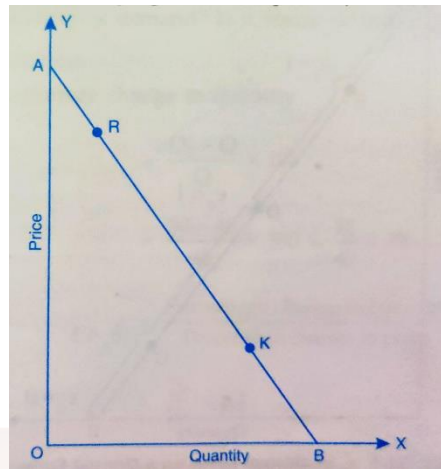
Hence $e_p > 1$ (Since $RB > RA$)

Similarly, if elasticity has to be measured at any other point on the demand curve. Say at point K

$$e_p \text{ at K} = \frac{KB}{KA}$$

Hence $e_p < 1$ (Since $KB < KA$)

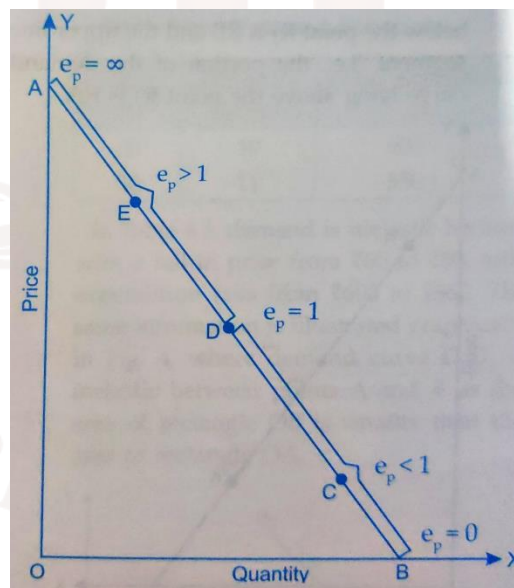
Therefore, this proves that price elasticity at different points on the demand curve is different.



Graph 4 E_p on straight line demand curve - Point method Graph

Source: Economics by DK Sethi.

Similarly, we can calculate the elasticity on various points:



Graph 5 showing E_p at different points on the demand curve

Source: DK Sethi Economics

AB is a straight-line demand curve, with A as its intercept on Y axis and B as X axis intercept and D as its mid-point. The point method of measuring elasticity as the ratio of line segment below the point and line segment above the points are used to illustrate elasticity at various points.

1. At point A = demand curve touches the vertical axis

$$e_p \text{ at point R} = \frac{\text{Line segment below A}}{\text{Line segment above A}}$$

$$= \frac{AB}{0} = \text{infinity}$$

2. At point above the midpoint but below A, say E

$$e_p \text{ at E} = \frac{BE}{EA} > 1$$

As the lower segment is greater than the upper segment.

3. At the mid-point D

$$e_p \text{ at D} = \frac{BD}{DA} = 1$$

As the lower segment equals the upper segment.

4. At any point below, mid-point but above B say at C

$$e_p \text{ at C} = \frac{BC}{CA} < 1$$

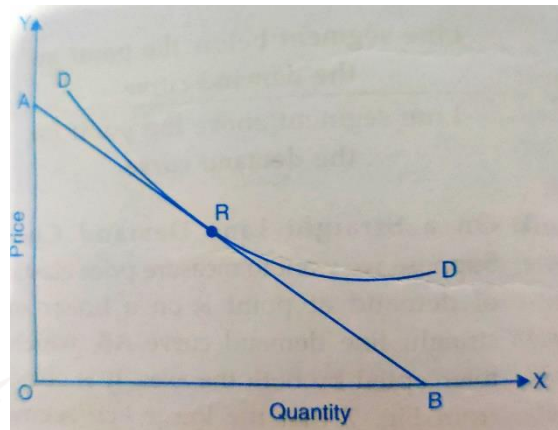
As the lower segment is smaller than the upper segment

5. At point B where the demand curve touches the horizontal axis

$$e_p \text{ at B} = \frac{0}{AB} = 0$$

Therefore, as we travel downwards from left to right on a straight-line demand curve, price elasticity of demand steadily varies from infinity at Y intercept to zero at X intercept being greater than unity (elastic demand) at any point above the midpoint, equal to the unitary at the mid-point and less than unity or inelastic demand at any point below the mid-point. From this we analyse that the curve is elastic towards the left-hand end and less elastic towards the right-hand end.

2. **On a Non-Linear demand curve:** Price elasticity of demand can be measured on a curve which is not a straight line by using point method. We draw a tangent to the demand curve through the chosen point and measure the elasticity of the tangent at this point as the ratio of the lower line segment to the upper line segment.



Graph 6 showing PE on non-linear demand curve

Source: Economics DK Sethi

If we want to measure price elasticity at point R on the demand curve DD as shown in the graph. We draw a line AB tangent to the demand curve DD at point R. Since the slope of the demand curve at R equals the slope of the tangent at that point, the price elasticity of the demand curve DD at R will be equal to the elasticity of the tangent AN=B at point R.

$$e_p \text{ at point } R = \frac{\text{Lower Line segment}}{\text{Upper Line segment}}$$

$$= \frac{BR}{RA}$$

Arc method

This method is suggested to measure large changes in both price and demand. When the price elasticity is to be found between the two prices, the question arises which price and quantity should be considered as the base. As elasticities found by using original price and quantity figures that serve as the base will be different from the one derived by using new price and quantity figures. Therefore, to avoid confusion generally the average of the two prices and quantities are taken as the base. When elasticity is measured over an interval of a demand curve, the elasticity is called as an interval or arc elasticity. It is the average elasticity over a segment or range of the demand curve. Hence, it is also called average elasticity of demand.

Formula:

$$= \frac{\Delta Q}{\Delta P} \times \frac{(P_1 + P_2)}{(Q_1 + Q_2)}$$

P1 = Original price

P2 = New price

Q1 = Original quantity

Q2 = New quantity

ΔQ = change in quantity

ΔP = change in price

For instance: A health and well-being store has a special offer on Hair wash bottles. It reduces their prices from Rs150 to Rs100. Suppose the store manager observes that the quantity demanded has increased from 700 bottles to 1300 bottles. Calculate the price elasticity of demand on the hair wash bottles

Formula:

$$= \frac{\Delta Q}{\Delta P} \times \frac{(P_1 + P_2)}{(Q_1 + Q_2)}$$

$$\Delta Q = 1300 - 700 = 600 \text{ (increased quantity - Original quantity)}$$

$$\Delta P = 150 - 100 = 50 \text{ (Old price - new price)}$$

$$P_1 = 150, P_2 = 100$$

$$Q_1 = 700, Q_2 = 1300$$

$$= 600 \times (150 - 100)$$

$$50 (700 + 1300)$$

$$= \frac{600}{50} \times \frac{250}{2000} = \frac{3}{2}$$

= 1.5 hence the hair wash bottle elasticity is 1.5

Proportionate or Percentage Method: Here elasticity is measured by the ratio between the proportionate or percentage change in the quantity demanded and percentage change in price.

$$E_p = \frac{\text{Percentage change in demand}}{\text{Percentage change in price}}$$

In order to find out percentage change in demand, the formula is –

$$\frac{\text{Change in quantity demand}}{\text{Original quantity demanded}} \times 100$$

Percentage or Proportionate Method

In this method, *price elasticity of demand* is measured by the ratio of percentage change in the quantity demanded to a percentage change in the price of the commodity. Thus:

$$e_p = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

$$= \frac{\frac{\text{Change in Quantity Demanded}}{\text{Initial Quantity}} \times 100}{\frac{\text{Change in Price}}{\text{Initial Price}} \times 100}$$

$$= \frac{\frac{\Delta Q}{Q} \times 100}{\frac{\Delta P}{P} \times 100} = \frac{\Delta Q}{Q} \div \frac{\Delta P}{P}$$

$$= \frac{\Delta Q}{Q} \times \frac{P}{\Delta P} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Thus:

$$e_p = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Where e_p stands for price elasticity of demand.

Q stands for initial quantity

P stands for initial price

ΔQ stands for change in quantity

ΔP stands for change in price.

Figure showing formula expansion.

Source Economics DK Sethi

For example

Price of A (Rs)	Quantity demanded of A (units)
5	10
4	15

When price of A is Rs 5 quantity demanded is 10 units and When Price falls to Rs.4 quantity demanded rises to 15 units.

$$\Delta P = \text{Rs } 5 - \text{Rs } 4 = \text{Rs } 1$$

$$\Delta Q = 15 - 10 = 5 \text{ units}$$

$$\text{Initial Price} = \text{Rs } 5$$

$$\text{Initial Quantity} = 10 \text{ units}$$

$$= \frac{P}{Q} \times \frac{\Delta q}{\Delta p} = \frac{5}{10} \times \frac{5}{1} = \frac{25}{10} = 2.5$$

Since price elasticity greater than one is Relatively elastic then the Elasticity of Product A is More than one which is 2.5 therefore this is Relatively elastic.

Numerical on Elasticity of demand:

Source Economics DK Sethi

Problem 6: The price elasticity of demand is 2. If price changes by 5 per cent. What will be the percentage change in quantity?

Solution:

$$e_p = \frac{\% \text{ Change in quantity demanded}}{\% \text{ Change in price}}$$

$$\Rightarrow 2 = \frac{\% \text{ Change in quantity demanded}}{5}$$

Hence, % change in quantity demanded = $2 \times 5 = 10$

Problem 7: The price elasticity of demand is 0.5. The % change in quantity is 6. What is % change in price?

Solution:

$$e_p = \frac{\% \text{ Change in quantity demanded}}{\% \text{ Change in price}}$$

$$\Rightarrow 0.5 = \frac{6}{\% \text{ Change in price}}$$

$$\Rightarrow \% \text{ Change in price} = \frac{6}{0.5} = \frac{6}{5/10} = 6 \times \frac{10}{5} = 12$$

Problem 8: Price elasticity of demand of a good is (-)1. At a given price the consumer buys 60 units of the good. How many units will the consumer buy if price falls by 10 per cent?

Solution:

$$e_p = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}}$$

$$\Rightarrow -1 = \frac{\% \text{ change in quantity demanded}}{-10\%}$$

$$\Rightarrow \% \text{ change in quantity demanded} = 10\%$$

$$\text{Change in demand} = 60 \times \frac{10}{100} = 6$$

$\therefore$ After the fall in price, demand will increase to = Initial quantity + change in quantity = $60 + 6 = 66$ units.

Problem 9: A fall in the price of a commodity by ₹2 per unit leads to an increase in quantity demanded by 10 units. Its price elasticity is (-)1. Calculate the quantity demanded at the initial price (before change) which was ₹10 per unit.

Solution:

Here, $\Delta P = -2$, $\Delta Q = 10$, $P = 10$, $e_p = (-1)$

$$e_p = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$$-1 = \frac{10}{-2} \times \frac{10}{Q} \text{ or } -1 = \frac{100}{-2Q}$$

$$\text{or } 2Q = 100$$

$$Q = 50$$

$\therefore$ Quantity demanded before change in price = 50 units.

Problem 10: 100 units of a good are demanded at the price of ₹4 per unit. The price rises and as a result its demand falls to 75 units. The price elasticity of demand of that good is (-)1. Find out the new price.

Solution:

Here, $P = 4$, $Q = 100$, $\Delta Q = 75 - 100 = -25$ (demand falls by 25), $e_p = (-)1$

$$e_p = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$$\Rightarrow -1 = \frac{-25}{\Delta P} \times \frac{4}{100} = \frac{-1}{\Delta P}$$

$$\Rightarrow \Delta P = 1$$

$\therefore$ New price = old price + increase in price = $4 + 1 = ₹5$ per unit.

Problem 11: A consumer buys 160 units of a good at price of ₹8 per unit. Price falls to ₹6 per unit. How much quantity will the consumer buy at the new price if price elasticity of demand is (-)2?

Solution:

Here, $P = 8$, $Q = 160$, $\Delta P = 6 - 8 = -2$ (price falls by ₹2 per unit), $e_p = (-)2$

$$e_p = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

$$\Rightarrow (-)2 = \frac{\Delta Q}{-2} \times \frac{8}{160}$$

$$\Rightarrow (-)2 = \frac{\Delta Q}{-40}$$

$$\Rightarrow \Delta Q = (-2) \times (-40) = 80$$

The consumer will buy 240 (160 + 80) units at the new price of ₹6 per unit.

Income elasticity of demand

Income elasticity of demand may be defined as the ratio or percentage change in the quantity demanded of a commodity to a given percentage change in the income. In short, it indicates the extent to which demand changes with a variation in consumer income. The following formula helps to measure E_y .

$$E_y = \frac{\text{Percentage change in demand}}{\text{Percentage change in income}}$$

Symbolically

$$E_y = \frac{\Delta D}{\Delta Y} \times \frac{Y}{D}$$

$$\frac{300}{2000} \times \frac{4000}{400} = 1.5$$

Original demand = 400 units Original Income = 4000-00 New demand = 700 units New Income = 6000-00

Generally speaking, E_y is positive. This is because there is a direct relationship between income and demand, i.e., the higher the income; higher would be the demand and vice versa. On the basis of the numerical value of the co-efficient, E_y is classified as greater than one, less than one, equal to one, equal to zero, and negative. The concept of E_y helps us in classifying commodities into different categories. Based on the value of E_y , the commodities can be classified as:

1. When E_y is positive, the commodity is normal [used in day-to-day life]
2. When E_y is negative, the commodity is inferior, e.g., Jowar, beedi, etc.
3. When E_y is positive and greater than one, the commodity is luxury.
4. When E_y is positive, but less than one, the commodity is essential.
5. When E_y is zero, the commodity is neutral, e.g. salt, match-box, etc.

Cross elasticity of demand

Cross elasticity demand may be defined as the percentage change in the quantity demanded of a particular commodity in response to a change in the price of another related commodity. In the words of Prof. Watson, cross elasticity of demand is the percentage change in quantity associated with a percentage change in the price of related goods. Generally, it arises in the case of substitutes and complements. The formula for calculating cross elasticity of demand is as follows:

$$E_c = \frac{\text{Percentage change in quantity demanded of commodity X}}{\text{Percentage change in the price of commodity Y}} = \frac{\Delta(Q_x \times P_y)}{(\Delta Q_x \Delta P_y)}$$

E_c stands for cross elasticity of demand of X for Y

Q_x stands for the original quantity demanded of X

ΔQ_x stands for change in quantity demanded of good X

P_y stands for the original price of good Y

ΔP_y stands for a small change in the price of good Y

Cross elasticity of demand measures how the quantity demanded of one good changes in response to a price change in another good. It helps identify whether goods are **complements** or **substitutes**.

Complementary Goods

Complementary goods are products that are used together, such as cars and fuel or coffee and sugar. A decrease in the price of one good increase the demand for its complement, and vice versa. In cross elasticity terms, complementary goods have a **negative cross elasticity**. For example, if the price of coffee decreases, people tend to buy more coffee, increasing the demand for sugar as well.

Substitution Goods

Substitute goods are products that can replace each other, such as tea and coffee or butter and margarine. An increase in the price of one good leads to an increase in the demand for its substitute. These goods have a **positive cross elasticity**. For example, if the price of coffee rises, consumers may switch to tea, increasing the demand for tea.

Let us take an example. If the price of coffee rises from Rs. 4.50 per hundred grams to Rs. 5 per hundred grams and as a result the consumer's demand for tea increases from 60 hundred grams to 70 hundred grams, then the cross elasticity of demand of tea for coffee can be found out as follows.

$$\Delta q_x = 70 - 60 = 10 \text{ hundred grams}$$

$$q_x = 60 \text{ hundred grams}$$

$$\Delta p_y = \text{Rs. } 5 - 4.50 = 50 \text{ paise}$$

$$p_y = \text{Rs. } 4.50 = 450 \text{ paise}$$

$$\begin{aligned} \text{Cross elasticity of demand} &= \frac{\Delta q_x}{\Delta p_y} \cdot \frac{p_y}{q_x} \\ &= \frac{10}{50} \times \frac{450}{60} = \frac{3}{2} = 1.5 \end{aligned}$$

It should be noted that:

1. Cross elasticity of demand is positive in case of good substitutes, e.g., coffee and tea.
2. High cross elasticity of demand exists for those commodities which are close substitutes. In other words, if commodities are perfect substitutes, for e.g., Bata or Corona shoes, Close-up or Pepsodent toothpaste, beans and ladies' finger, Pepsi and Coca-Cola, etc.

3. The cross elasticity is zero when commodities are independent of each other, for e.g., stainless steel, aluminium vessels, etc.
4. Cross elasticity between two goods is negative when they are complements. In these cases, a rise in the price of one will lead to a fall in the quantity demanded of another commodity, for example, car and petrol, pen and ink, etc.

Advertising or promotional elasticity of demand

Most of the firms, in the present marketing conditions, spend considerable amounts of money on advertising and other such sales promotional activities with the object of promoting their sales. Advertising elasticity refers to the responsiveness of demand or sales to change in advertising or other promotional expenses. The formula to calculate the advertising elasticity is as follows:

$$E_a = \frac{\text{Percentage change in demand or sales}}{\text{Percentage change in Advertisement expenditure}}$$

Symbolically,

$$E_a = \frac{\Delta D \text{ or Sales}}{A \times A \text{ Demand or sales}}$$

$$\frac{40,000}{1200 \times 800 / 10,000} = 2.67$$

Original sales = 10,000 units original advertisement expenditure = 800-00 New sales = 50,000 units new advertisement expenditure = 2000-00

In the above example, advertising elasticity of demand is 2.67. It implies that for every one-time increase in advertising expenditure, the sales would go up 2.67 times. Thus, E_a is more than one.

SELF-ASSESSMENT QUESTIONS – 2

Fill in the blanks:	
7	Law of demand explains the _____ change in demand and elasticity of demand explains _____ change in demand.
8	According to Marshall, _____ is the degree of responsiveness of demand to the change in price of that commodity.
9	The relatively elastic demand curve is _____.

10	When the quantity demanded increases with the increase in income, we say that income elasticity of demand will be _____. When quantity demanded decreases with an increase in income, we say that the income elasticity of demand is _____.
11	_____ helps the manager to decide the advertisement expense.
12	Point method helps to measure _____ quantity of change in demand and arc methods helps to measure _____ changes in demand.
True or False Questions	
vi)	A good faces inelastic demand if the quantity demanded increases significantly when the price decreases slightly.
vii)	Goods that have close substitutes have more elastic demand than goods that do not have close substitutes.
viii)	The demand curve is vertical and elasticity is 0 when demand is perfectly inelastic.
ix)	Cross-price elasticity can be used to determine if goods are inferior or normal goods.
x)	The elasticity of a linear, downward-sloping demand curve is constant while its slope is not constant.



4. SUMMARY

Let us recapitulate the important concepts discussed in this unit:

- Demand is created by consumers. Consumers can create demand only when they have adequate purchasing power and willingness to buy different goods and services. There is a direct relationship between utility and demand. The law of demand tells us that there is an inverse relationship between price and demand in general. Sometimes, customers buy more in spite of a rise in the prices of some commodities.

Thus, the law of demand has certain exceptions. Demand for a product not only depends on price but also on a number of other factors. In order to know the quantitative changes in both price and demand, one has to study elasticity of demand.

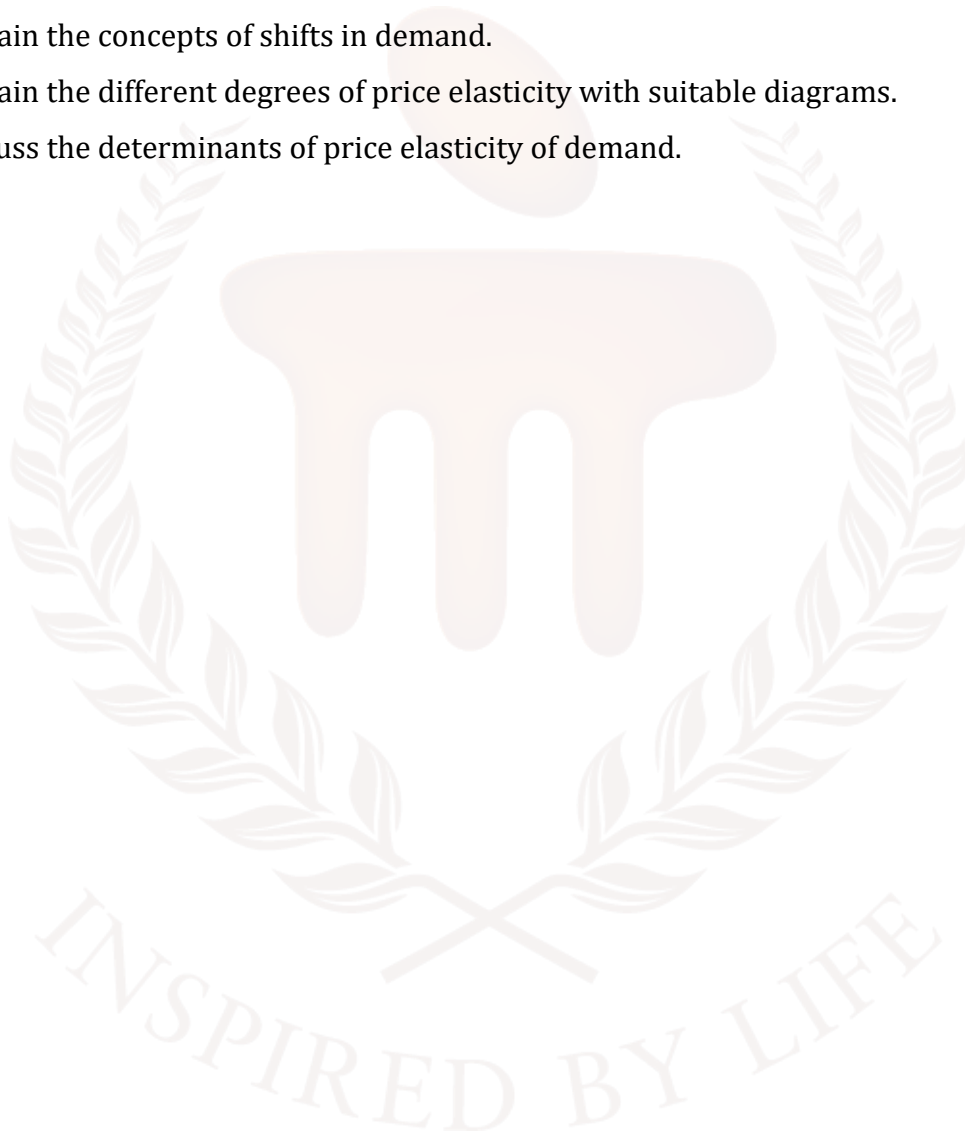
- Price elasticity of demand indicates the percentage changes in demand as a consequence of changes in prices. The response of demand to price changes is different. Hence, we have elastic and inelastic demand. One can measure exactly the extent of price elasticity of demand with the help of different methods like point and arc methods. Income elasticity measures the quantum of changes in demand in response to changes in income of the customers.
- Cross elasticity tells us the extent of a change in the price of one commodity and corresponding changes in the demand for another related commodity. Substitution elasticity measures the amount of changes in demand ratio of two substitute goods to changes in price ratio of two substitute goods in the market. The concept of elasticity of demand has great theoretical and practical application in all aspects of business life.

5. GLOSSARY

Demand	- It is the total or given quantity of a commodity or a service that is purchased by the consumer in the market at a particular price and at a particular time.
Demand curve	- A locus of points showing various alternative price-quantity combinations.
Demand function	- A comprehensive formulation which specifies the factors that influence the demand for a product.
Elasticity of demand	- Responsiveness or sensitivity of demand to a given change in the price or non-price determinant of a commodity.
Law of Demand	- Keeping other factors that affect demand constant, a fall in price of a product leads to an increase in quantity demanded and a rise in price leads to decrease in quantity demanded for the product.
Necessaries	- Items which are purchased by consumers whatever the price may be.
Speculation	- Purchase or sale of an asset with the hope that its price may rise or fall and make speculative profit.
Veblen's effect	- Demand for status symbol goods would go up with a rise in price and vice versa.

6. TERMINAL QUESTIONS

1. State and explain the law of demand.
2. Discuss the various exceptions to the law of demand.
3. Explain the concepts of shifts in demand.
4. Explain the different degrees of price elasticity with suitable diagrams.
5. Discuss the determinants of price elasticity of demand.



7. ANSWERS

Self-Assessment Questions

1. Inversely
2. Same / upward
3. Expansion, contraction
4. Qualitative
5. Comprehensive / wider
6. Fall
7. Direction, percentage
8. Price Elasticity of Demand
9. Flatter
10. Positive; negative
11. Advertisement Elasticity of Demand
12. Small, large

True or False

- i) False
- ii) True
- iii) True
- iv) False
- v) False
- vi) False
- vii) True
- viii) True
- ix) False
- x) False

Terminal Questions

1. The term demand refers to total or given quantity of a commodity or a service that is purchased by the consumer in the market at a particular price and at a particular time. Refer to Section 2.2.
2. Exceptions to the law of demand state that with a fall in price, demand also falls and with a rise in price demand also rises. Refer to Section 2.2 for more details.
3. If demand increases, there will be a forward shift in the demand curve to the right and if demand decreases, then there will be a backward shift in the demand curve. Refer to Section 2.2.
4. Elasticity of demand shows the reaction of one variable with respect to a change in other variables on which it is dependent. Refer to Section 2.3.
5. The elasticity of demand depends on several factors Refer to Sections 2.3.

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